Chapter 7: Continuous Probability Distributions

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Introduction

In this chapter, we study situations where the random variable X is a continuous number and so it is drawn from a continuous probability distribution.

In order for you to calculate what values X can take, you must examine it across a range within the distribution.

Uniform Distribution

Consider an example where a passenger is waiting for a train. The passenger knows that the train always arrives at the station in 20 minutes, but they do not know how long it has been since it last arrived.

We are interested in the probability of how long a passenger must wait for the bus to arrive.

Uniform Distribution

Let the random variable X be defined as **the number of minutes a passenger must wait** until the train arrives:

- The maximum number of minutes a passenger must wait is 20.
- The minimum number of minutes a passenger must wait is 0.

Uniform Distribution

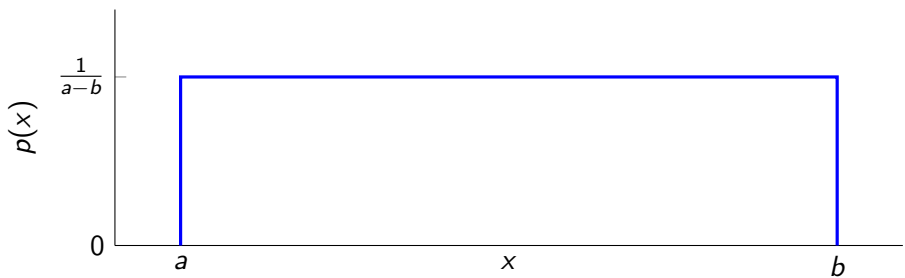
When a continuous random variable X has a lower bound a and upper bound b and that the probability of the event is equally likely on all values of X, then we say that **X** is **uniformly distributed**:

$$X \sim U(a, b).$$

Uniform Distribution

We can illustrate this idea on a graph:

Figure 1: Uniform Distribution of Variable X



Uniform Distribution

Here, the rectangle is a **uniform probability distribution**.

Notice how the probability that any value of X is all the same. It is represented by the height of the rectangle, $\frac{1}{a-b}$. The length of the rectangle is the distance between b and a.

Uniform Distribution

The bus stop scenario is an example of a random variable which is drawn from a uniform distribution.

This is because when a passenger arrives at the station, they do not know when the train will arrive, and only that it always arrives in 20 minutes since it last left.

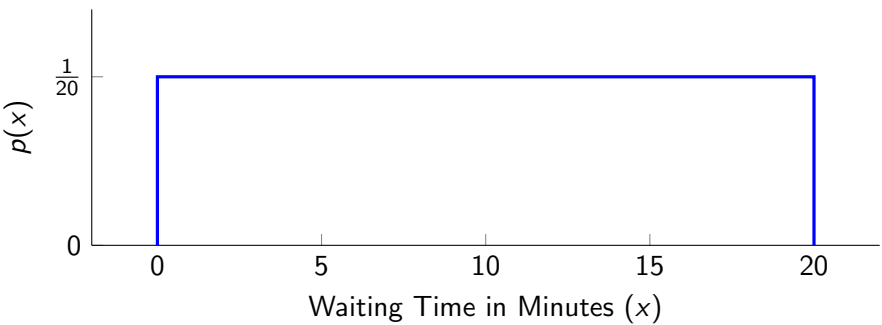
The probability that the train arrives is equal for any number between 0 and 20 minutes.

$$X \sim U(0, 20)$$

Uniform Distribution

We can illustrate this on a graph:

Figure 2: Uniform Distribution of Passenger Waiting Time



Uniform Distribution

Suppose the passenger arrives at the station and wants to know the probability that they need to wait less than 10 minutes for the train to arrive. Formally, we are trying to find:

$$P(X < 10).$$

To do this, we are essentially trying to find the area of the rectangle (Remember that the area of a rectangle is Width x Height.) between 0 and 10:

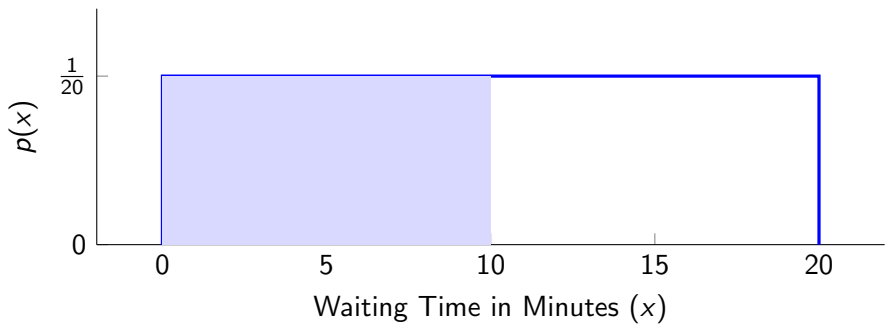
Uniform Distribution

$$P(X < 10) = (10 - 0) \cdot \left(\frac{1}{20} - 0 \right)$$
$$= \frac{10}{20} = 0.5$$

that is, the probability that they wait less than 10 minutes is 50%.

Uniform Distribution

Figure 3: Uniform Distribution of Passenger Waiting Time



Uniform Distribution

Activity

Find the probability that the passenger waits:

(a) Less than 5 minutes.

(b) Less than 20 minutes.

Uniform Distribution

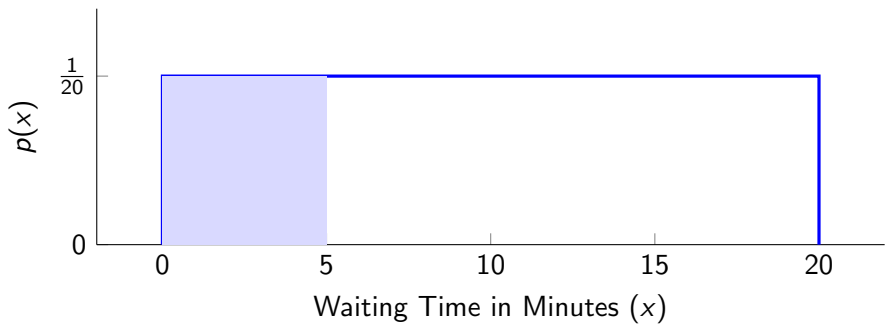
Solution: The probability that the passenger waits less than 5 minutes is:

$$P(X < 5) = (5 - 0) \cdot \left(\frac{1}{20} - 0 \right)$$
$$= 0.25.$$

There is a 25% chance they have to wait less than 5 minutes:

Uniform Distribution

Figure 4: Uniform Distribution of Passenger Waiting Time



Uniform Distribution

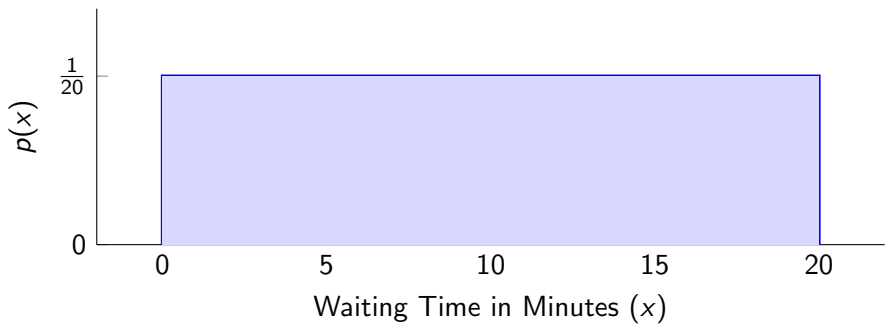
The probability that the passenger waits less than 20 minutes is:

$$P(X < 20) = (20 - 0) \cdot \left(\frac{1}{20} - 0 \right)$$
$$= 1.$$

There is a 100% chance the passenger has to wait less than 20 minutes. This is true given that the train always arrives in 20 minutes.

Uniform Distribution

Figure 5: Uniform Distribution of Passenger Waiting Time



Uniform Distribution

Before leaving this section, you need to be aware of a technical aspect when we deal with continuous variables. In particular, we cannot acquire a probability outcome when try to calculate only one exact value of X. For example, the probability that a passenger waits exactly 10 minutes is:

$$P(X = 10) = 0$$

Uniform Distribution

This is because of the nature of continuous variables. In theory, the random variable X can take a infinite number of values. For example, the number of minutes a person waits is actually 10.559232312 minutes.

Uniform Distribution

As a result, whenever we analyse continuous variables, we always specify it within particular ranges:

$$P(X < 10)$$
$$P(10 < X < 20)$$
$$P(X > 15)$$

that way, the entire range of values are covered within the density of the probability distribution.

Normal Distribution

Previously, we saw that when a continuous variable X is generated from a uniform distribution, then the probability of seeing a value is equally likely.

We now turn towards the most famous kind of continuous variable distribution in statistics.

Normal Distribution

Suppose we generate the random variable X from a distribution with the following observations:

1. Most values tend to be densely concentrated around the mean.
2. Less values tend to be away from the mean.

then this is the characteristic of a **normal distribution**.

Normal Distribution

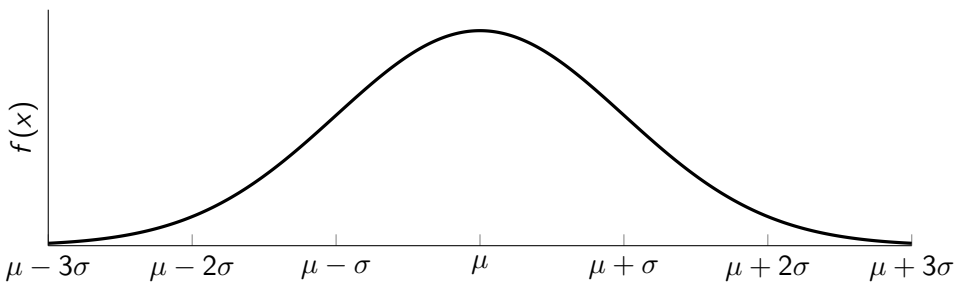
Formally, we say that X is generated from a normal distribution with mean μ and variance σ^2:

$$X \sim N(\mu, \sigma^2)$$

Normal Distribution

We can illustrate the normal distribution with the canonical symmetric bell shape:

Figure 6: The Normal Distribution



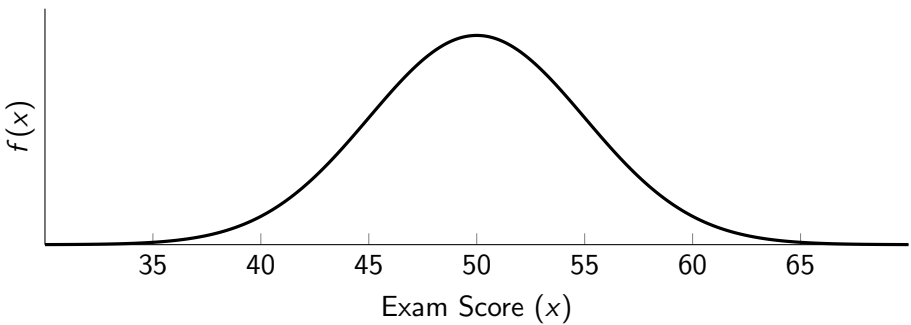
Normal Distribution

Consider a population of students where X represents the mathematics exam score of a randomly selected student. Suppose that exam scores are normally distributed with a mean of 50 and a variance of 25:

$$\mu = 50 \qquad \sigma^2 = 25$$

Normal Distribution

Figure 7: The Normal Distribution



Normal Distribution

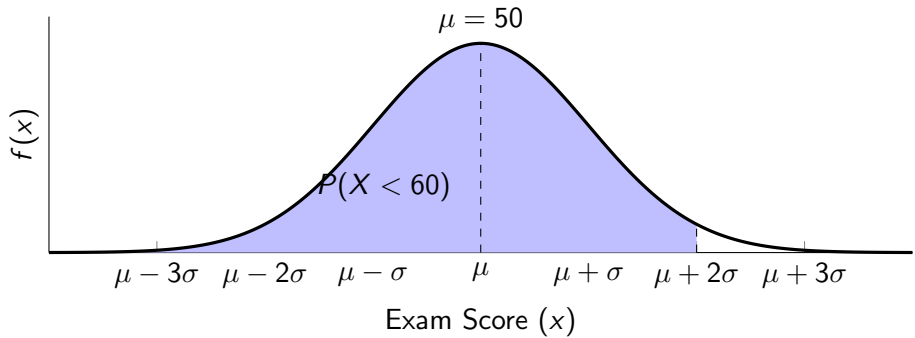
Just like we did before, we can use this information to understand the probability of observing a particular event. For example, we are interested in the probability that we observe a student that scored less than 60 in the exam:

$$P(X < 60)$$

The probability of observing this is represented by the shaded area of the normal distribution:

Normal Distribution

Figure 8: The Normal Distribution



The probability $P(X < 60)$ is therefore represented by the area under the normal density curve to the left of $X = 60$.

Normal Distribution

However, unlike the uniform distribution, however, calculating this area is not straightforward. The normal distribution is non-linear, so we cannot simply calculate the area using the formula for a rectangle.

To calculate normal probabilities, we can instead **standardise** the values of X and use the **standard normal distribution**.

Standardising the Normal Distribution

In order to calculate the area underneath the curve in an effective way, statisticians tend to **standardise** the normal distribution.

This means that no matter what X is, so long as it is normally distributed, then we can calculate the probabilities of observing an event in the same way.

We define the following formula:

$$z = \frac{x - \mu}{\sigma}$$

Standardising the Normal Distribution

When you input the values of x, μ and σ, then you will be able to measure how many **standard deviations** a particular value of x is away from the mean.

Standardising the Normal Distribution

Activity:

Continuing from our exam score example:

$$X \sim N(50, 25)$$

Using the z formula, calculate how many standard deviations away the following exam scores are from the mean:

Standardising the Normal Distribution

(a) $x = 60$
(b) $x = 43$
(c) $x = 50$

Standardising the Normal Distribution

Solution:

Using the formula:

(a) When $x = 60$:

$$z = \frac{60 - 50}{5} = 2$$

The score of 60 is 2 standard deviations away from the mean.

Standardising the Normal Distribution

(b) When $x = 43$:

$$z = \frac{43 - 50}{5} = -1.4$$

The score of 43 is -1.4 standard deviations away from the mean.

Standardising the Normal Distribution

(c) When $x = 50$:

$$z = \frac{50 - 50}{5} = 0$$

The score of 50 is 0 standard deviations away from the mean.

Standardising the Normal Distribution

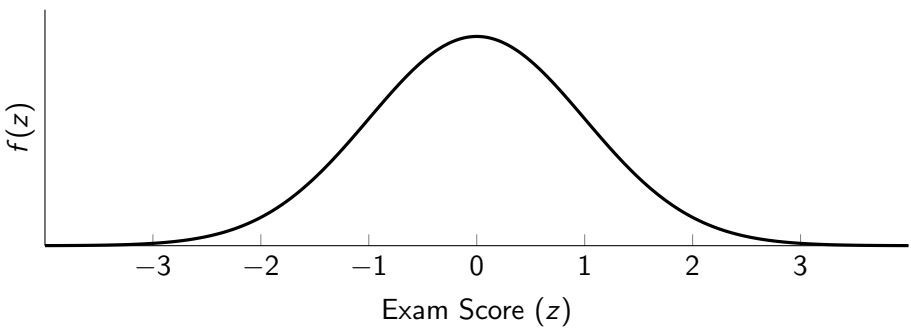
Because of this, we don't actually need to list out the values of exam scores on the horizontal axis. So long as the data follows from a normal distribution, we can just standardise the horizontal axis. Formally, the **standard normal distribution** is defined as:

$$X \sim Z(0, 1)$$

That is, Z has a mean $\mu = 0$ and a variance $\sigma^2 = 1$. Note that the standard deviation is the square root of the variance, and that is also just $\sigma = 1$.

Standardising the Normal Distribution

Figure 9: The Standard Normal Distribution



Case 1

Now that we have standardised the horizontal axis, we can start with learning how to calculate the probability of observing values of the random variable X.

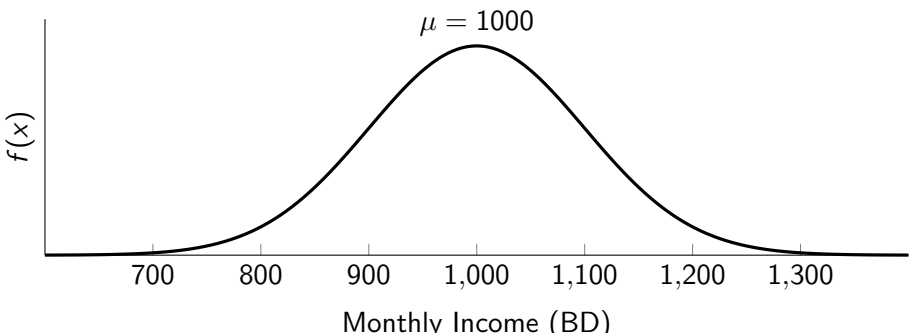
Consider if you have data on the income of all Bahrainis. You discover that the average monthly income $\mu = 1000$ and its variance $\sigma^2 = 10,000$:

$$X \sim N(1000, 10000)$$

which implies that $\sigma = 100$:

Case 1

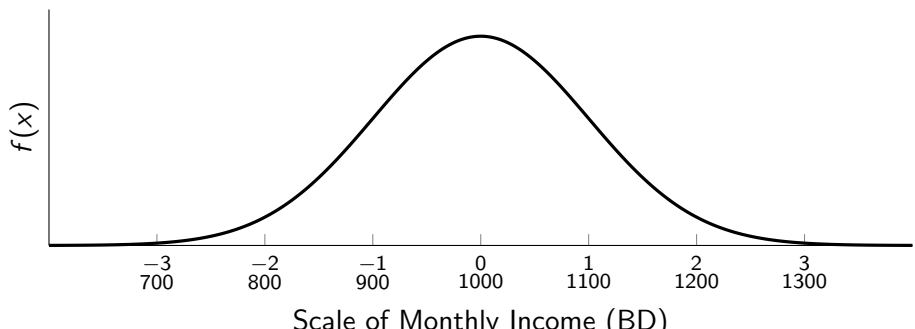
Figure 10: Normal Distribution of Monthly Income



Case 1

In order to easily calculate the probability, we instead work with the standardised values of X where we just measure how far away it is from the mean:

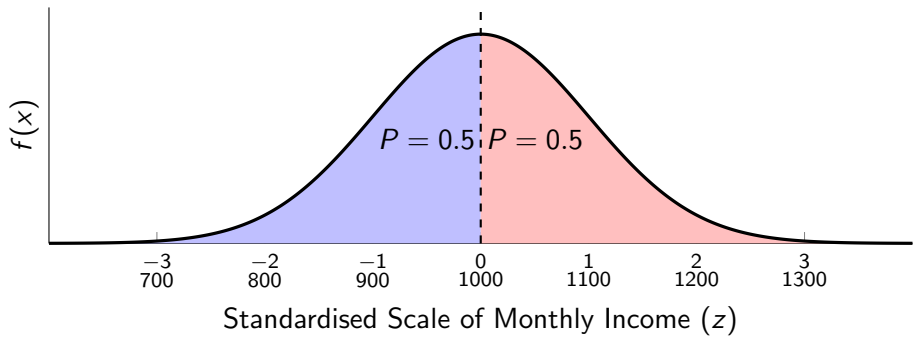
Figure 11: Standard Normal Distribution of Income



Case 1

Let us start with a basic fact, in that we say **half** of the observations are located **under the mean** while the other **half** are located **above the mean**:

Figure 12: Standard Normal Distribution of Income



Case 1

Let us start with a basic exercise and calculate how many standard deviations away a person is that earns 1000 BD and another person earning 1100 BD:

$$z = \frac{1000 - 1000}{100} = 0$$

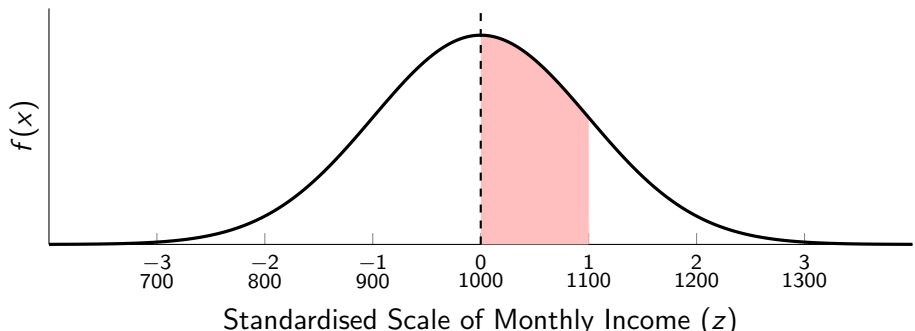
$$z = \frac{1100 - 1000}{100} = 1$$

that is, a person earning 1000 BD is 0 standard deviations away from the mean, whereas a person earning 1,100 BD is 1 standard deviation away from the mean.

Case 1

From the population, a researcher is interested in determining the probability that they observe someone who has an income **between** 1000 and 1100. Graphically, this is simplt the shaded region:

Figure 13: Standard Normal Distribution of Income



Case 1

To answer the question, you would need to calculate the area of the red region, however this is not straight forward given that this is a non-linear function and you would need to rely on calculus.

However given that the values of X has been standardised, then so long as the distribution is normal, we can calculate the area of any region using the z-table (SEE HANDOUT).

Case 1

What each of these values represent is the area in which:

$$P(0 < Z < z)$$

In the case of finding the probability that a person earns between 1000 and 1100 BD, we are simply asking:

$$P(1000 < X < 1100) = P(0 < Z < 1)$$

Case 1

To answer this, look at the first column and find the value for $z = 1$ under the column $z = 0.00$:

Table 1: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749

Case 1

What this value means is that the probability of observing a value between $z = 0$ and $z = 1$ is 0.3413. Formally:

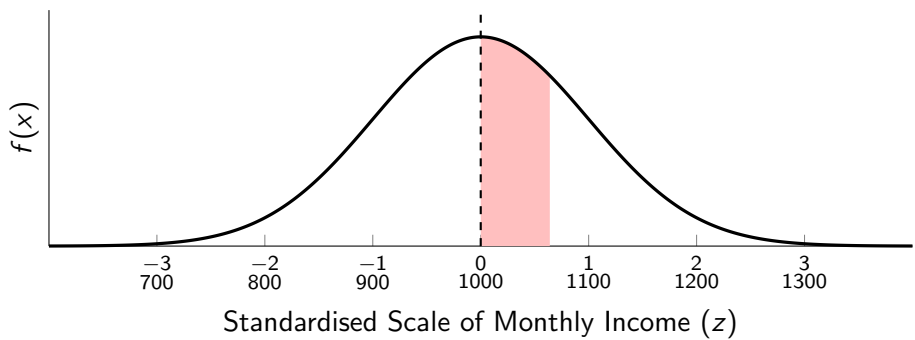
$$P(1000 < X < 1100) = P(0 < Z < 1) = 0.3413$$

which is 34.13%.

Case 1

To understand why we only look at the first column at 0.00, let us consider another example where we try to determine the probability that a person earns between 1000 BD and 1064 BD:

Figure 14: Standard Normal Distribution of Income



Case 1

The z values are:

$$z = \frac{1000 - 1000}{100} = 0$$
$$z = \frac{1064 - 1000}{100} = 0.64$$

Therefore, we want the probability that $P(1000 < X < 1064) = P(0 < Z < 0.64)$. Notice how in the table we do not have the exact value for $z = 0.64$. To determine the probability, we need to find the row where $z = 0.6$ and the column that says 0.04:

Case 1

Table 2: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734

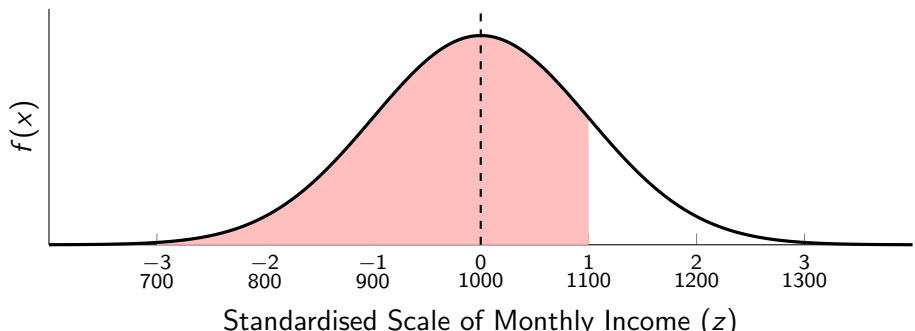
Case 1

Therefore $P(0 < Z < 0.64) = 0.2389$ or 23.89%. This is how you would know which **row** and **column** to use for each case.

Case 2

Now suppose we are interested in determining the probability that a person earns less than 1100 BD per month. Graphically, this is simply:

Figure 15: Standard Normal Distribution of Income



Case 2

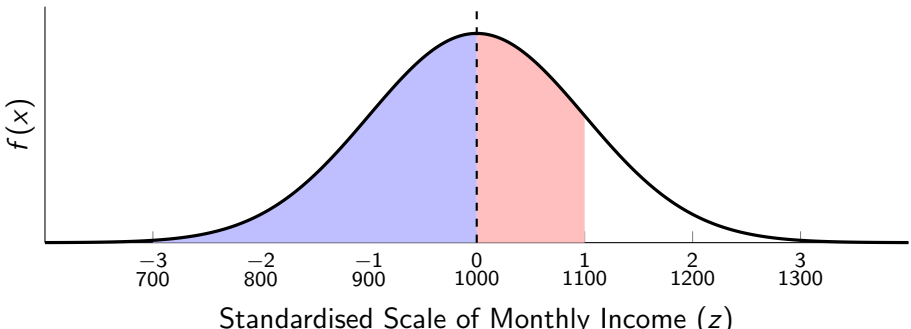
The problem is that the z-table only gives you the probability values for when the lower bound is always fixed at $z = 0$. Therefore, you cannot look at the table and find $P(X < 1100) = P(Z < 1)$.

However there is a way around this. Remember that we said earlier that all values less than the mean represent half of the observations.

Intuitively, this means that the probability of observing **anyone** with $z < 0$ is simply $P = 0.5$:

Case 2

Figure 16: Standard Normal Distribution of Income



where the blue region represents $P(X < 1000) = P(Z < 0) = 0.5$.

Case 2

We also know what the red region is:

$$z = \frac{1100 - 1000}{100} = 1$$

which is $P(1000 < X < 1100) = P(0 < Z < 1) = 0.3413$.

Case 2

Therefore, the probability of observing someone that earns less than 1100 BD is:

$$= P(X < 1000) + P(1000 < X < 1100)$$

$$= P(Z < 0) + P(0 < Z < 1)$$

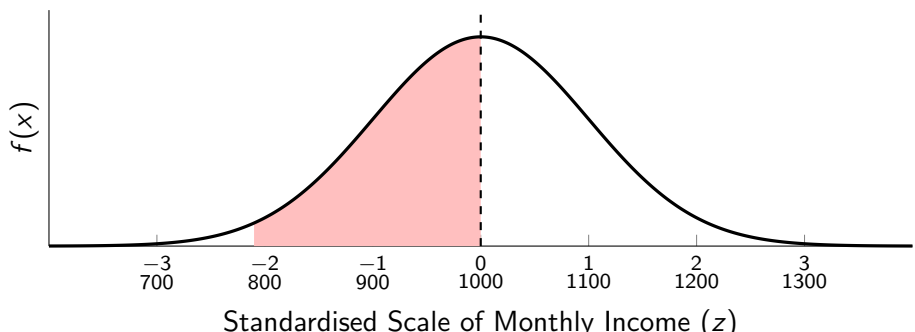
$$= 0.5 + 0.3413 = 0.8413$$

which is 84.13%.

Case 3

Now suppose we want to calculate the probability that a person earns between 790 BD and 1000 BD. This is similar to what we did in Case 1, but this time the value of z will be below 0:

Figure 17: Standard Normal Distribution of Income



Case 3

Let us get the values of z:

$$z = \frac{790 - 1000}{100} = -2.1$$
$$z = \frac{1000 - 1000}{100} = 0$$

Remember that the table shows probability values for $P(0 < Z < z)$. The problem is that $z = -2.1$ is **less** than 0.

Case 3

However, we can easily get around this by understanding that the normal distribution is **symmetric** around the mean.

This means that all the probability values are identical for both positive and negative values of z.

Case 3

Therefore, to find the probability that someone earns between 790 and 1000 BD, simply take the **absolute value** (ignore the negative sign) of the z:

$$|z| = 2.1$$

Case 3

When looking at the table:

Table 3: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878

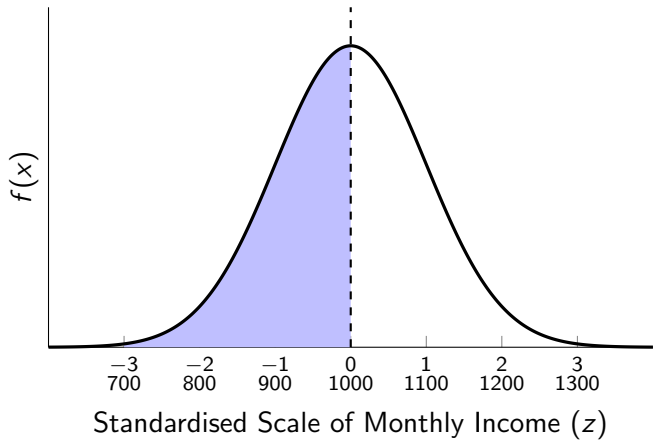
Case 3

Therefore, the probability of observing someone between 790 and 1000 BD is $P(790 < X < 1000) = 0.4821$.

Case 4

To understand how to solve this next case, remember that the probability of observing any value below the mean is simply $p = 0.5$ if the distribution is normal:

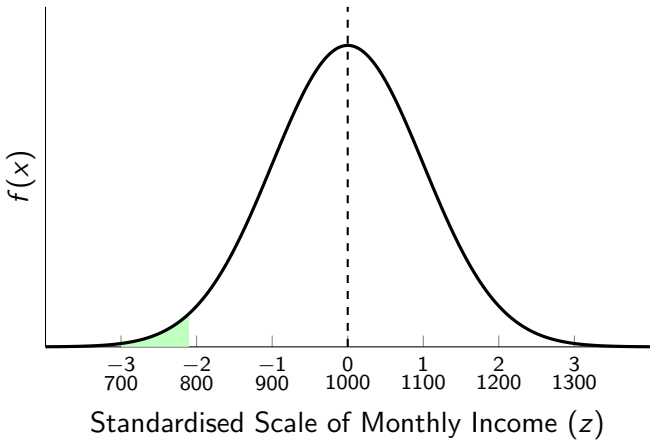
Figure 18: Standard Normal Distribution of Income



Case 4

Suppose now we want to know the probability that we observe someone that earns less than 790 BD $P(X < 790)$. Graphically this is simply:

Figure 19: Standard Normal Distribution of Income



Case 4

Once again, you cannot get this area from the table since it identifies the calculations only with a lower bound and an upper bound rather than just $P(X < 790)$.

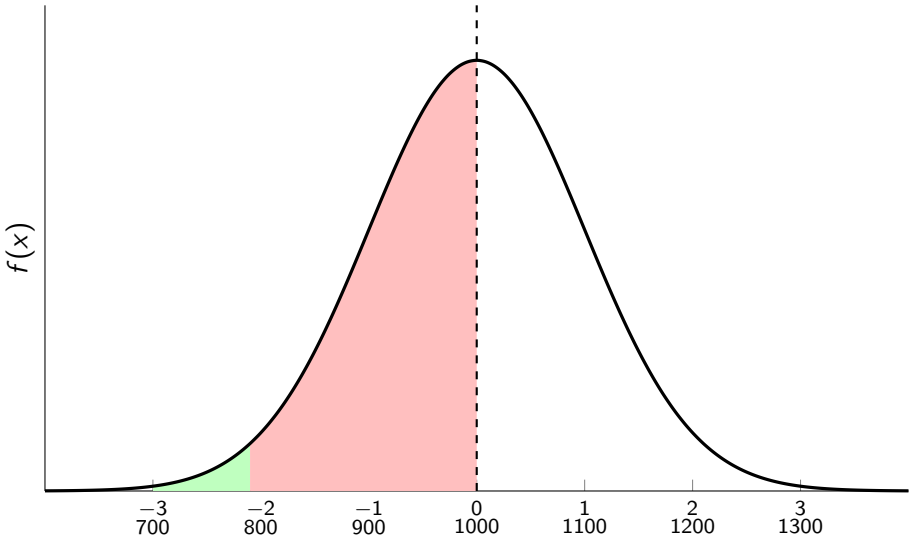
This means that if you were to calculate the z:

$$z = \frac{790 - 1000}{100} = -2.1$$

and so $|z| = 2.1$, then this is the area between 0 and 2.1:

Case 4

Figure 20: Standard Normal Distribution of Income



Case 4

which we know is $P(0 < Z < 2.1) = 0.4821$:

Table 4: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878

Case 4

Using this information, it is possible to determine the area of the green region. Since the probability of observing any number below the mean is $P(X < 1000) = P(Z < 0) = 0.5$, then it must mean that the green region is:

$$\begin{aligned} & P(Z < 0) - P(0 < Z < 2.1) \\ & = 0.5 - 0.48921 \\ & = 0.01079, \end{aligned}$$

which means you have a 1.079% chance of observing a Bahraini who earns less than 790 BD.

Case 5

Suppose we are now interested in the probability that we observe a Bahraini who earns between $P(845 < X < 1200)$. Let us first get the z values:

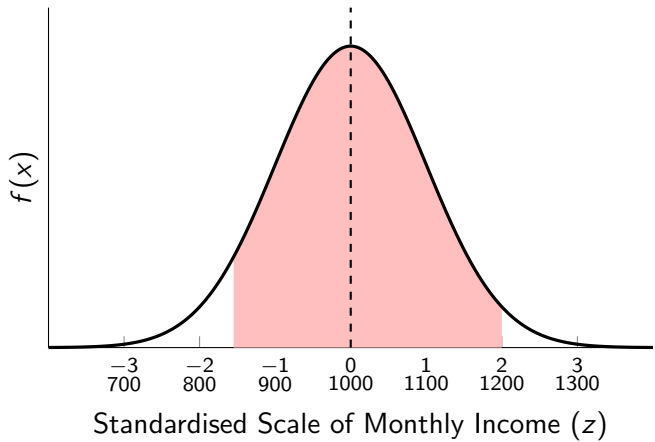
$$z = \frac{845 - 1000}{100} = -1.55$$

$$z = \frac{1200 - 1000}{100} = 2$$

Case 5

In the figure, we are interested in the probability of the regions between $z = -1.55$ and $z = 2$:

Figure 21: Standard Normal Distribution of Income



Case 5

Remember that since the distribution is symmetric, we do not need to concern ourselves with the negative sign, and so we can just find for $|z| = 1.55$ and $z = 2$:

Table 5: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842

Case 5

Therefore the probability of observing someone who earns between 890 BD and 1200 BD is:

$$= P(890 < X < 1200)$$

$$= P(0 < Z < 1.55) + P(0 < Z < 2)$$

$$= 0.4394 + 0.4772$$

$$= 0.9166$$

which is a 91.66% probability.

Case 6

Now consider the probability of observing a Bahraini who earns between $P(1150 < X < 1215)$. Let us start by getting the z values:

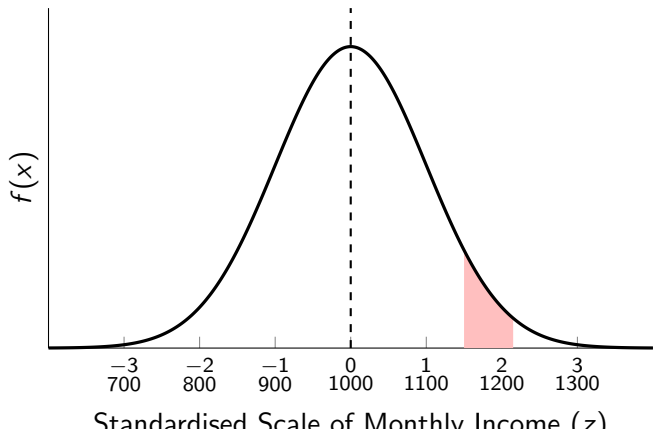
$$z = \frac{1150 - 1000}{100} = 1.5$$

$$z = \frac{1215 - 1000}{100} = 2.15$$

Case 6

Graphically:

Figure 22: Standard Normal Distribution of Income



Case 6

Now remember that the problem with the z table is that it shows you probability values where the lower bound is always 0 and the upper bound is only one number when you are analysing $P(0 < Z < z)$.

Therefore you cannot use the table on its own to reveal $P(1.5 < Z < 2.15)$. Instead, you can only get:

$$P(0 < Z < 1.5) = 0.4332$$

$$P(0 < Z < 2.15) = 0.4842$$

Case 6

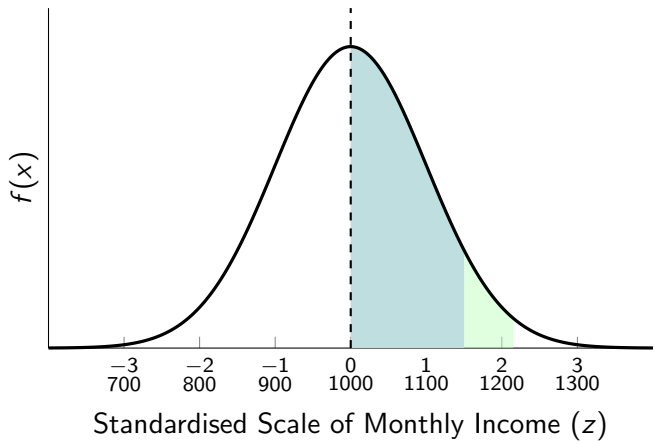
Table 6: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878

Case 6

Graphically, these regions overlap:

Figure 23: Standard Normal Distribution of Income



Case 6

You can acquire the value of the overlapping region by just subtracting the two probabilities:

$$= P(1000 < X < 1215) - P(1000 < X < 1150)$$

$$= P(0 < Z < 1.5) - P(0 < Z < 2.15)$$

$$= 0.4842 - 0.4332$$

$$= 0.0510$$

hence the area of the red shaded region represents a probability of 5.1%.

Case 7

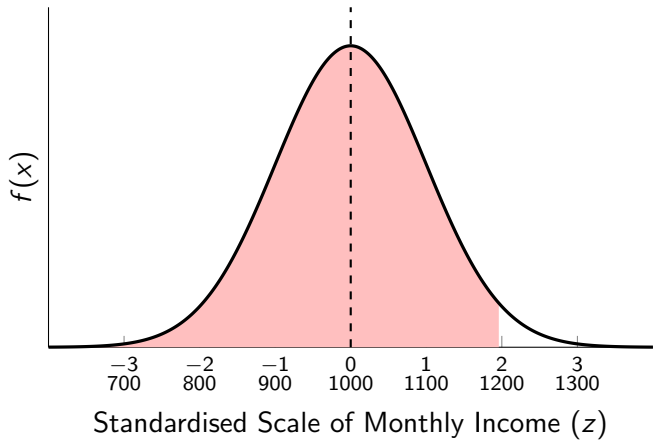
Before going through this case, you must remember the concept of a percentile. Recall that the *kth* percentile of a dataset is the location to which $k\%$ of data are below it.For example:

1. The 95th percentile is the value of X such that 95% of the data values are below it.
2. The 5th percentile is the value of X such that 5% of the data values are below it.

Case 7

For the Bahraini income example, suppose you are interested in knowing the value of x which is the 97.5th percentile of the data. Graphically, this looks like:

Figure 24: Standard Normal Distribution of Income



Case 7

In other words, we want the value of income x such that we have a 97.5% chance of observing it.

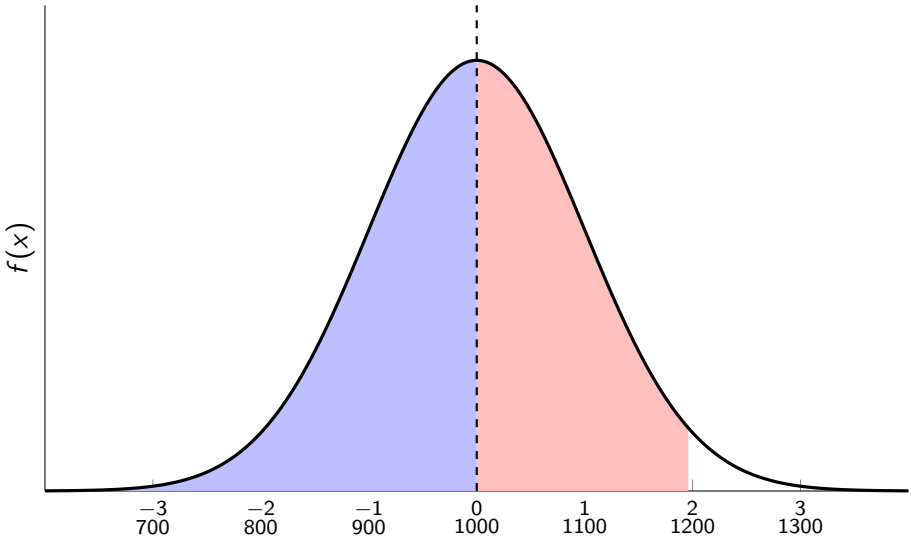
The graph shows you the region where $P(X < x) = 0.975$. We are interested in finding the value of this x.

Case 7

Let us start by finding the z value. Remember that table only shows you values for $P(0 < Z < z)$. We know that all the values below the mean represent $P = 0.5$:

Case 7

Figure 25: Standard Normal Distribution of Income



Case 7

Therefore:

$$= P(X < x) = 0.975$$

$$= P(X < 1000) + P(1000 < X < x) = 0.975$$

$$= P(Z < 0) + P(0 < Z < z) = 0.975$$

$$= 0.5 + P(0 < Z < z) = 0.975$$

$$= P(0 < Z < z) = 0.475$$

Case 7

We can now find the value of z that gives us $P(0 < Z < z) = 0.475$:

Table 7: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.48033

Therefore, $P(0 < Z < z) = 0.475$ only when $z = 1.96$.

Case 7

Now that we know the value of z which gives us the 97.5th percentile of x, we can substitute it in the z formula:

$$z = \frac{x - \mu}{\sigma} = 1.96$$

$$= \frac{x - 1000}{100} = 1.96$$

$$x = 1196$$

therefore, A Bahraini earning $x = 1196$ is the 97.5th percentile of the data.